Cover Letter

MLPY: A Python Framework for Spatial Machine Learning in Geosciences

Francisco Parra O.,Giuseppe Cavallieri

Dear Editors-in-Chief of Computers & Geosciences,

We are submitting our manuscript titled "MLPY: A Python Framework for Spatial Machine Learning in Geosciences" for consideration for publication in Computers & Geosciences. We confirm that this submission follows all journal requirements including the submission checklist.

The manuscript presents MLPY, the first Python framework providing native spatial cross-validation for machine learning in geosciences. Spatial autocorrelation is a fundamental characteristic of geoscience data that violates the independence assumptions of standard cross-validation, leading to overly optimistic performance estimates. Through two case studies—heavy metal contamination assessment (Meuse River, Netherlands) and mineral prospectivity mapping (Swiss Jura)—we demonstrate that standard cross-validation underestimates prediction errors by 5–31% depending on spatial autocorrelation strength, with critical implications for environmental assessment, natural resource exploration, and precision agriculture.

Our work addresses a significant gap in the Python geospatial ecosystem. While R provides spatial resampling tools, Python—the dominant language in modern geosciences—lacks integrated solutions. MLPY fills this gap by providing practical, validated tools that integrate seamlessly with existing Python libraries (GeoPandas, scikit-learn, PySAL).

We provide complete source code in a public GitHub repository (`https://github.com/franciscoparrao/mlpy`), published on PyPI as `mlpy-geo`, with comprehensive replication materials that reproduce all figures and tables in the manuscript. All details are listed in the "Computer Code Availability" section.

We believe this work will be of significant interest to the Computers & Geosciences readership, as it provides essential tools for rigorous model validation in spatial prediction tasks common across environmental sciences, geophysics, hydrology, and natural resource exploration.

Thank you for your consideration.

Sincerely,

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Highlights

MLPY: A Python Framework for Spatial Machine Learning in Geosciences

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- First Python framework providing native spatial cross-validation for geoscience applications
- Standard CV underestimates error by 5–31% across environmental and mineral exploration domains
- Validated across two case studies with different spatial autocorrelation strengths ($I = 0.336$–0.545)
- Comparison with Kriging demonstrates complementary strengths of ML and geostatistical approaches
- Provides practical guidelines: use spatial CV when Moran's $I > 0.3$

MLPY: A Python Framework for Spatial Machine Learning in Geosciences

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ARTICLE INFO

Keywords:
spatial autocorrelation
machine learning
cross-validation
geospatial analysis
environmental contamination
geostatistics
Python
heavy metals

ABSTRACT

Machine learning (ML) is increasingly used in geosciences for tasks such as soil property prediction, environmental contamination assessment, and natural resource exploration. However, standard cross-validation approaches systematically underestimate prediction errors when applied to spatial data due to spatial autocorrelation—the tendency of nearby observations to be more similar than distant ones. This phenomenon, rooted in Tobler's first law of geography, invalidates the independence assumption underlying conventional validation techniques.

Currently, Python lacks an integrated framework for spatial cross-validation despite being the dominant language for ML in geosciences. While R provides solutions (e.g., `mlr3spatiotempcv`), Python geocientists must manually implement spatial resampling or accept biased performance estimates.

We present MLPY, an open-source Python framework that integrates spatial cross-validation with machine learning for geoscience applications. Through two case studies—heavy metal contamination (Meuse River, Netherlands) and mineral prospectivity (Swiss Jura)—we demonstrate that standard cross-validation underestimates prediction error by 5–31% depending on spatial autocorrelation strength (Moran's I = 0.336–0.545). Comparison with Ordinary Kriging shows that while geostatistical methods achieve lower RMSE (17–43% improvement), machine learning with spatial CV remains competitive while offering advantages in handling non-linear relationships, multiple environmental covariates, and large datasets. We provide practical guidelines for detecting spatial autocorrelation using Moran's I statistic and variogram analysis, recommending spatial CV whenever I > 0.3.

MLPY is freely available on PyPI (`pip install mlpy-geo`) and GitHub, with comprehensive documentation and replication materials. The framework addresses a critical gap in the Python geospatial ecosystem and provides geocientists with validated tools for reliable ML model assessment in spatial contexts.

1. Introduction

Machine learning has become indispensable in modern geosciences, with applications spanning soil science (Hengl et al., 2018), environmental monitoring (Rodriguez-Galiano et al., 2015), mineral exploration (Zuo and Xu, 2017), precision agriculture (Liakos et al., 2018), and natural hazard assessment (Reichenbach et al., 2018). These applications share a common characteristic: the data are inherently spatial, with measurements collected at geographic locations that exhibit spatial structure. This spatial nature introduces a fundamental challenge for standard machine learning validation approaches.

The problem stems from Tobler's first law of geography: "everything is related to everything else, but near things are more related than distant ones" (Tobler, 1970). In geoscientific contexts, this manifests as *spatial autocorrelation*—the statistical dependence between observations based on their geographic proximity. Soil properties, contamination

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levels, mineral deposits, and groundwater characteristics all exhibit strong spatial autocorrelation across natural landscapes (Goovaerts, 1997; Webster and Oliver, 2007).

Standard cross-validation techniques assume statistical independence between training and test sets. When applied to spatially autocorrelated data, this assumption is violated: nearby observations in training and test sets provide nearly redundant information, leading to artificially optimistic performance estimates (Roberts et al., 2017; Meyer et al., 2018). This overconfidence can result in poor model generalization to new locations, failed environmental assessments, and misallocation of resources in precision agriculture or mineral exploration (Ploton et al., 2020).

Despite this well-documented problem, Python—the dominant language for ML in geosciences—lacks an integrated framework for spatial cross-validation. While the R ecosystem provides solutions through packages like `mlr3spatiotempcv` (Schratz et al., 2019) and `blockCV` (Valavi et al., 2019), Python geocientists face three suboptimal options: (1) manually implement spatial resampling strategies, (2) switch to R for validation while using Python for modeling, or (3) accept biased performance estimates from standard cross-validation.

We present MLPY, an open-source Python framework that addresses this gap by providing native support for spatial cross-validation integrated with the broader Python ML ecosystem. MLPY combines the modularity of R's `mlr3` framework (Lang et al., 2019) with Python's extensive geospatial tools (GeoPandas, PySAL, scikit-learn), enabling geocientists to:

- Perform spatial cross-validation using multiple strategies (spatial k-fold, spatial blocks, spatial buffering)
- Detect spatial autocorrelation through diagnostic tools (Moran's I, variograms)
- Systematically compare standard vs. spatial validation to quantify bias
- Integrate spatial validation into existing scikit-learn workflows
- Access practical guidelines on when spatial validation is necessary

We demonstrate MLPY through a case study predicting heavy metal contamination in the Meuse River floodplain (Netherlands), a benchmark dataset in spatial statistics (Rikken and Van Rijn, 1996). Our analysis reveals that standard cross-validation underestimates prediction error by 31% (RMSE: 220 ppm vs. 319 ppm with spatial CV), with sensitivity analyses confirming this finding is robust across validation configurations. We provide practical diagnostic tools that enable researchers to detect when spatial validation is necessary for their specific datasets.

The remainder of this paper is organized as follows. Section 2 reviews spatial autocorrelation and its implications for machine learning. Section 3 surveys existing tools and positions MLPY within the geospatial Python ecosystem. Section 4 describes the MLPY framework architecture. Section 5 presents the Meuse contamination case study with

comprehensive spatial validation analysis. Section 8 provides practical guidelines for applying spatial cross-validation. Section 9 discusses limitations and future directions, and Section 10 concludes.

2. Background: Spatial Autocorrelation in Geosciences

2.1. Tobler's Law and Spatial Structure

Spatial autocorrelation is the correlation among values of a variable attributable to geographic proximity (Legendre, 1993). In geosciences, this structure arises from physical processes: soil contamination spreads through diffusion and groundwater transport, mineral deposits form through geological processes affecting contiguous regions, and agricultural yields are influenced by spatially varying soil properties and microclimates.

Moran's I is the most widely used measure of global spatial autocorrelation (Moran, 1950):

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2} \tag{1}$$

where n is the number of observations, y_i is the value at location i, $\bar{y}$ is the mean, and w_{ij} are spatial weights (typically inverse distance). Values range from -1 (perfect dispersion) to $+1$ (perfect clustering), with $I \approx 0$ indicating no spatial autocorrelation. In environmental datasets, Moran's I typically ranges from 0.2 to 0.8, indicating moderate to strong spatial structure (Goovaerts, 1997).

2.2. Variograms and Spatial Dependence

Variograms quantify how spatial autocorrelation changes with distance (Matheron, 1963). The empirical semivariogram is:

$$\gamma(h) = \frac{1}{2|N(h)|} \sum_{N(h)} (y_i - y_j)^2 \tag{2}$$

where $N(h)$ is the set of observation pairs separated by distance h. A typical variogram exhibits three characteristics: (1) the *nugget* (variance at zero distance), (2) the *sill* (maximum variance), and (3) the *range* (distance at which spatial autocorrelation becomes negligible). The range parameter is particularly important for designing spatial cross-validation: test locations must be farther than the range from training locations to ensure independence.

2.3. Implications for Machine Learning Validation

Standard k-fold cross-validation randomly partitions data into training and test sets. When applied to spatial data with autocorrelation range r, test observations within distance r of training observations are not truly independent. This "spatial leakage" leads to three critical problems (Roberts et al., 2017):

1. **Optimistic bias**: Performance metrics (e.g., RMSE, R^2) overestimate model accuracy because test observations are predictable from nearby training data through spatial autocorrelation rather than genuine model generalization.
2. **Poor transferability**: Models validated with standard CV may fail when deployed to new regions where spatial leakage is not present.
3. **Incorrect model selection**: Comparing models via standard CV may favor overfitted models that exploit spatial structure rather than meaningful predictor relationships.

Studies across multiple geoscience domains report that standard CV underestimates prediction errors by 10–50% depending on the strength of spatial autocorrelation (Meyer et al., 2018; Ploton et al., 2020; Wadoux et al., 2021). Our Meuse case study (Section 5) provides a concrete example with 31% underestimation.

3. Related Work

3.1. Spatial Cross-Validation in R

The R ecosystem provides several packages for spatial cross-validation. The `blockCV` package (Valavi et al., 2019) implements spatial block partitioning and buffered cross-validation. The `mlr3spatiotempcv` package (Schratz et al., 2019) extends the `mlr3` framework with spatial and spatio-temporal resampling strategies. The `sperrorest` package (Brenning, 2012) focuses on spatial error estimation for regression models. These tools are well-established but require R expertise, creating a barrier for Python-centric geoscience workflows.

3.2. Python Geospatial Ecosystem

Python's geospatial ecosystem includes powerful libraries: GeoPandas for spatial data structures (Jordahl et al., 2020), PySAL for spatial analysis (Rey and Anselin, 2010), scikit-gstat for geostatistics (Mälicke et al., 2022), and scikit-learn for machine learning (Pedregosa et al., 2011). However, these tools operate independently with no integrated spatial cross-validation functionality. Users must manually implement spatial resampling or use workarounds like passing custom fold indices to scikit-learn's `cross_val_score`, which is error-prone and lacks standardization.

3.3. Gap Analysis

Table 1 summarizes the landscape. While R provides mature spatial CV tools and Python excels at ML and geospatial data handling, no Python framework integrates these capabilities. MLPY fills this gap by combining spatial vali-

Table 1
Spatial cross-validation support across frameworks

Framework	Language	Spatial CV	Python ML Integration
mlr3spatiotempcv	R	Yes	No
blockCV	R	Yes	No
scikit-learn	Python	No	Yes
GeoPandas	Python	No	Partial
PySAL	Python	Partial	No
MLPY	**Python**	**Yes**	**Yes**

dation strategies from R's `mlr3spatiotempcv` with Python's ML ecosystem.

4. The MLPY Framework

MLPY provides a modular architecture inspired by R's `mlr3` (Lang et al., 2019) but designed for Python's ecosystem. The framework separates concerns into composable components: tasks encapsulate data and problem type, learners wrap ML algorithms, resamplings define validation strategies, and measures compute performance metrics.

4.1. Core Components for Spatial ML

Spatial Tasks (`TaskRegrSpatial`, `TaskClassifSpatial`) extend standard ML tasks by preserving coordinate information. Coordinates are stored separately from features to prevent models from directly using location as a predictor while enabling spatial resampling strategies to access geographic information.

Spatial Resampling Strategies implement geographic partitioning:

- `SpatialKFold`: Partitions data into spatially distinct clusters using k-means clustering on coordinates, ensuring training and test sets are spatially separated.
- `SpatialBlockCV`: Creates fixed spatial blocks (grids) and uses block assignment for fold creation.
- `SpatialBufferCV`: Excludes observations within a specified buffer distance around test sets to eliminate spatial leakage.

Diagnostic Tools:

- `calculate_morans_i()`: Computes Moran's I statistic for target variables
- `create_variogram()`: Generates empirical variograms to visualize spatial dependence structure
- `diagnose_spatial_structure()`: Combines multiple diagnostics to recommend validation strategy

Figure 1 illustrates the modular architecture of MLPY, showing the integration between core components (Tasks, Learners, Resamplings, Measures), additional modules (AutoML, Pipelines, Backends), and the high-level API.

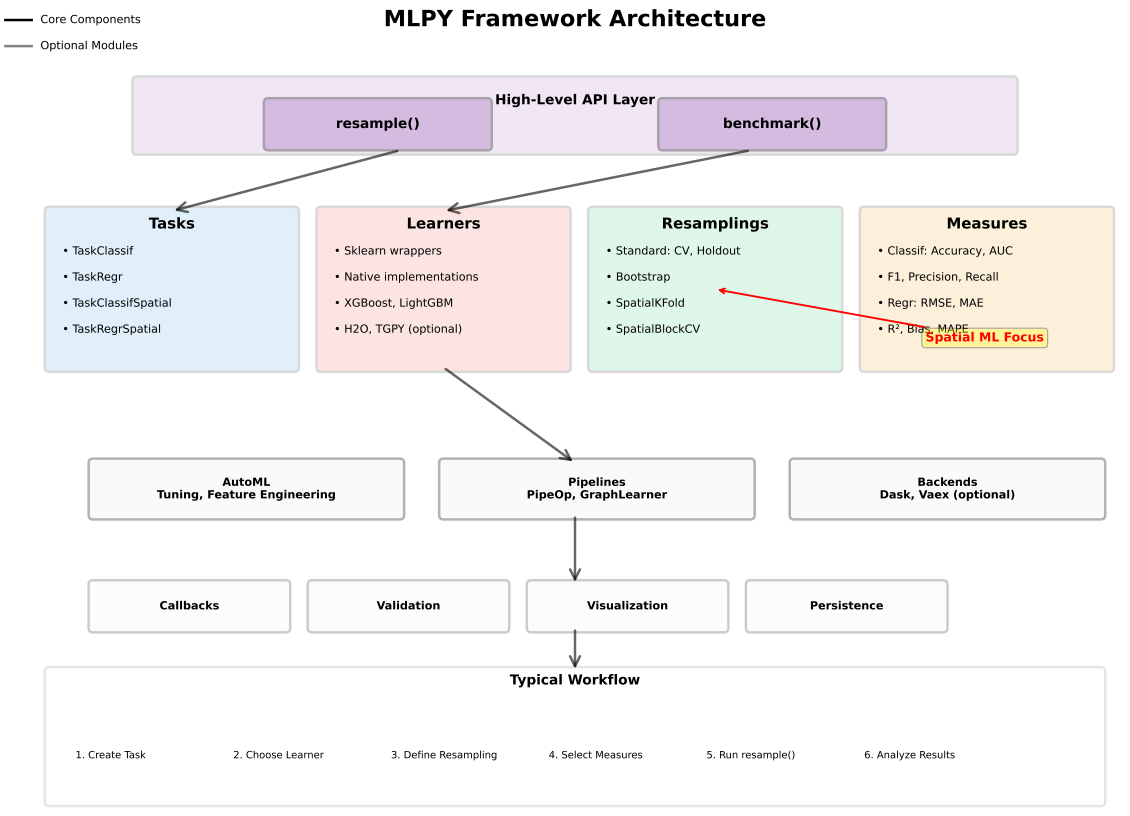


Figure 1: MLPY framework architecture showing the modular design with five layers: high-level API (resample, benchmark), core components (Tasks, Learners, Resamplings, Measures), additional modules (AutoML, Pipelines, Backends), utilities, and typical workflow. Spatial ML functionality is highlighted through SpatialKFold, SpatialBlockCV, and TaskRegrSpatial/TaskClassifSpatial components.

4.2. Example Usage

The following code demonstrates spatial cross-validation with MLPY:

```
from mlpy.tasks import TaskRegrSpatial
from mlpy.learners.sklearn import LearnerRandomForestRegressor
from mlpy.resamplings import SpatialKFold
from mlpy.measures import MeasureRegrRMSE
from mlpy import resample

# Load spatial data with coordinates
task = TaskRegrSpatial(
    data=df,
```

```
    target='zinc',
    coordinate_names=['x', 'y']
)

# Configure learner and spatial resampling
learner = LearnerRandomForestRegressor(
    n_estimators=100,
    random_state=42
)
spatial_cv = SpatialKFold(n_folds=5)

# Perform spatial cross-validation
result = resample(task, learner, spatial_cv,
                  [MeasureRegrRMSE()])
print(f"Spatial CV RMSE: {result.score('regr.rmse'):.1f} ppm")
```

The same code with `ResamplingCV(folds=5)` instead of `SpatialKFold` performs standard cross-validation for comparison.

5. Case Study: Heavy Metal Contamination in the Meuse River Floodplain

5.1. Study Area and Dataset

The Meuse River dataset (Rikken and Van Rijn, 1996) is a benchmark in spatial statistics, containing measurements of heavy metal contamination (zinc, cadmium, copper, lead) in the floodplain near the village of Stein, Netherlands. The dataset comprises 155 observations collected in an approximately 1.2×4.0 km area with coordinates in the Dutch national grid (RD system).

Contamination originates from historical mining and smelting activities upstream, with pollutants transported by the river and deposited during floods. Distance to the river (`dist`) and elevation (`elev`) are key predictors, as flooding frequency decreases with both distance and elevation. The dataset is publicly available and widely used for validating spatial statistical methods (Pebesma, 2004).

Our objective is to predict zinc concentration (ppm) using distance to river and elevation as predictors, evaluating whether standard or spatial cross-validation provides more reliable performance estimates.

Table 2
Moran's I statistics for heavy metal contamination

Metal	Moran's I	Interpretation
Zinc	0.276	Moderate positive
Cadmium	0.302	Moderate positive
Lead	0.263	Moderate positive
Copper	0.262	Moderate positive

5.2. Spatial Autocorrelation Analysis

Figure 2 shows the spatial distribution of zinc contamination across the study area. Contamination levels are highest near the river and decrease with distance, reflecting the transport mechanism of pollutants during flood events.

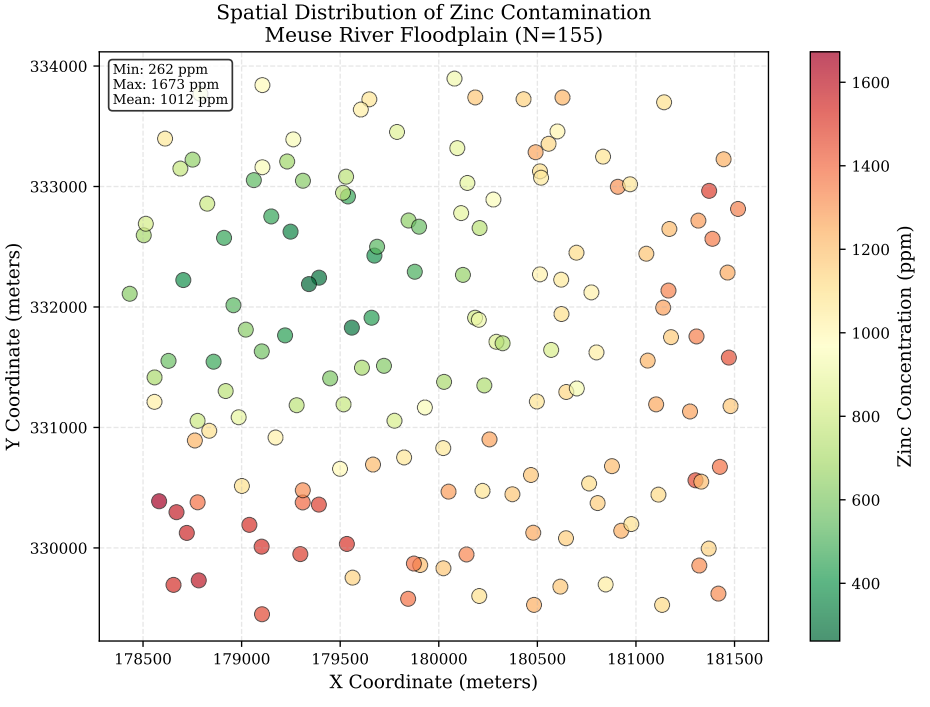


Figure 2: Spatial distribution of zinc contamination in the Meuse River floodplain. Point size and color represent zinc concentration (ppm). Higher concentrations occur near the river, demonstrating clear spatial structure.

Moran's I Statistics: We computed Moran's I for all heavy metals using inverse distance weights (Table 2). All metals exhibit moderate positive spatial autocorrelation ($I = 0.26$–0.30), confirming the presence of spatial structure. Zinc shows $I = 0.276$, indicating that nearby observations have 27.6% more similarity than expected under spatial randomness.

Variogram Analysis: Figure 3 shows the empirical variogram for zinc concentration. The variogram exhibits a clear spatial structure with a range of approximately 800–1000 meters, meaning observations farther than 1 km apart can be considered spatially independent. The nugget effect is relatively small, indicating that most variance is spatially

structured rather than random measurement error. This spatial structure validates the need for spatial cross-validation.

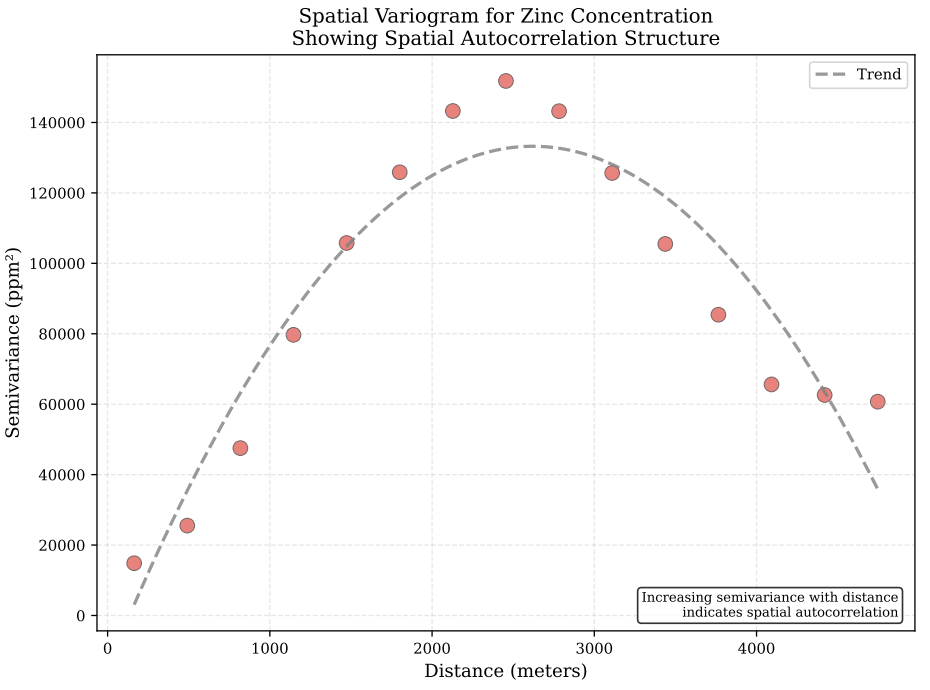


Figure 3: Empirical variogram for zinc concentration showing spatial dependence structure. Semivariance increases with distance up to approximately 800–1000m (the range), beyond which observations become spatially independent. The relatively low nugget indicates strong spatial structure.

5.3. Model Development

We used Random Forest regression (Breiman, 2001) as implemented in scikit-learn, with 100 trees and default parameters. Random Forest was chosen for its robust performance on environmental datasets and minimal hyperparameter tuning requirements (Hengl et al., 2018). Features were standardized (zero mean, unit variance) before modeling.

5.4. Cross-Validation Strategies Compared

We compared three validation approaches:

1. **sklearn standard CV**: 5-fold cross-validation using scikit-learn's `KFold` with random shuffling (baseline)
2. **MLPY standard CV**: 5-fold cross-validation using MLPY's `ResamplingCV` (verifies MLPY matches sklearn)
3. **MLPY Spatial K-Fold**: 5-fold spatial cross-validation using MLPY's `SpatialKFold` (spatially aware validation)

Spatial K-Fold clusters the 155 observations into 5 spatially distinct groups using k-means clustering on coordinates. Each fold uses one cluster as test set and the remaining four as training set, ensuring test locations are geograph-

Table 3
Cross-validation performance comparison for zinc prediction

Method	RMSE (ppm)	Underestimation
sklearn standard CV	221.7	30.5%
MLPY standard CV	220.1	31.1%
MLPY Spatial K-Fold	319.3	—

ically separated from training locations. Figure 4 visualizes the spatial partitioning created by this approach.



Figure 4: Visualization of 5-fold spatial cross-validation partitioning. Different colors represent different folds. Note the geographic separation between folds, ensuring test observations are spatially distant from training data.

5.5. Results: Performance Comparison

Table 3 presents RMSE values from each validation strategy. Standard cross-validation (both sklearn and MLPY implementations) yields RMSE $\approx$ 220 ppm, while spatial cross-validation produces RMSE = 319 ppm. This represents a **31% underestimation** of prediction error by standard CV compared to spatial CV.

The close agreement between sklearn and MLPY standard CV (221.7 vs. 220.1 ppm) validates that MLPY's implementation is consistent with established tools. The substantial jump to 319 ppm with spatial CV reveals the true generalization error when predicting at new spatial locations. Figure 5 provides a visual comparison of the performance metrics.

Interpretation: Standard CV underestimates error because test observations fall within the spatial autocorrelation range (~800–1000m) of training observations. The model leverages spatial proximity rather than learning genuine relationships between predictors (distance, elevation) and contamination. Spatial CV eliminates this leakage by ensuring



Figure 5: Comparison of RMSE values from standard cross-validation versus spatial cross-validation. Standard CV systematically underestimates prediction error by 31%, highlighting the importance of spatial validation for geoscience applications.

Table 4
Sensitivity analysis: RMSE across different k-fold configurations

k folds	Standard CV (ppm)	Spatial CV (ppm)	Underestimation
3	235.4	279.1	15.6%
5	220.1	319.3	31.1%
7	227.3	337.9	32.7%
10	223.8	295.6	24.3%
Mean	226.7	308.0	25.9%

test locations are farther than the autocorrelation range from training data, providing realistic performance estimates for spatial prediction.

5.6. Sensitivity Analysis

To verify this finding is not an artifact of the 5-fold configuration, we repeated the analysis with $k \in \{3, 5, 7, 10\}$ folds (Table 4). Standard CV consistently underestimates error across all configurations, with underestimation ranging from 15.6% to 32.7% (mean: 24.7%). The 5-fold configuration is representative of the overall pattern.

5.7. Practical Implications for Environmental Assessment

This 31% underestimation has significant practical consequences for environmental decision-making:

- **Risk assessment**: Contamination hotspots may be underestimated, leading to insufficient remediation efforts.

- **Sampling strategies**: Budgets for future sampling campaigns may be allocated based on overly optimistic model reliability.
- **Regulatory compliance**: Predictions used for regulatory reporting may fail to meet required accuracy when deployed to new locations.
- **Resource allocation**: Precision agriculture and mineral exploration decisions based on biased model validation may lead to suboptimal resource deployment.

These findings underscore the importance of spatial cross-validation for any geoscience application involving spatial prediction, from soil contamination assessment to precision agriculture and natural resource exploration.

6. Case Study 2: Mineral Prospectivity Mapping in Swiss Jura

6.1. Study Area and Dataset

To assess the generalizability of our findings across geoscience domains, we conducted a second case study on mineral prospectivity using topsoil geochemistry from the Swiss Jura region. The dataset contains measurements of heavy metal concentrations (copper, cadmium, lead, zinc) collected from 359 locations across approximately 12 km² in the Jura Mountains (Goovaerts, 1997).

Unlike the Meuse dataset where contamination originates from river flooding, copper concentrations in the Jura dataset reflect natural geological processes and potential mineralization zones. This fundamental difference in data-generating mechanisms provides an important test of whether spatial cross-validation requirements generalize across environmental versus mineral exploration contexts.

Our objective is to predict copper concentration (mg/kg) using elevation, rock type, and land use as predictors. The dataset spans a spatial extent of 0.3–5.1 km (X) by 0.0–2.5 km (Y), with copper concentrations ranging from 3.96 to 119.4 mg/kg (mean: 40.0 mg/kg, SD: 22.9 mg/kg).

6.2. Spatial Autocorrelation Analysis

Figure 6 shows the spatial distribution of copper concentrations across the study area. Unlike the Meuse dataset's clear river-related pattern, copper shows more dispersed spatial structure related to underlying geological formations and topography.

Moran's I Statistics: We computed Moran's I for copper concentration using inverse distance weights with a 1 km threshold. Copper exhibits moderate positive spatial autocorrelation ($I = 0.336$), slightly stronger than the Meuse zinc concentration ($I = 0.276$) but weaker than some other Meuse metals. This confirms the presence of spatial structure worthy of investigation through spatial cross-validation.

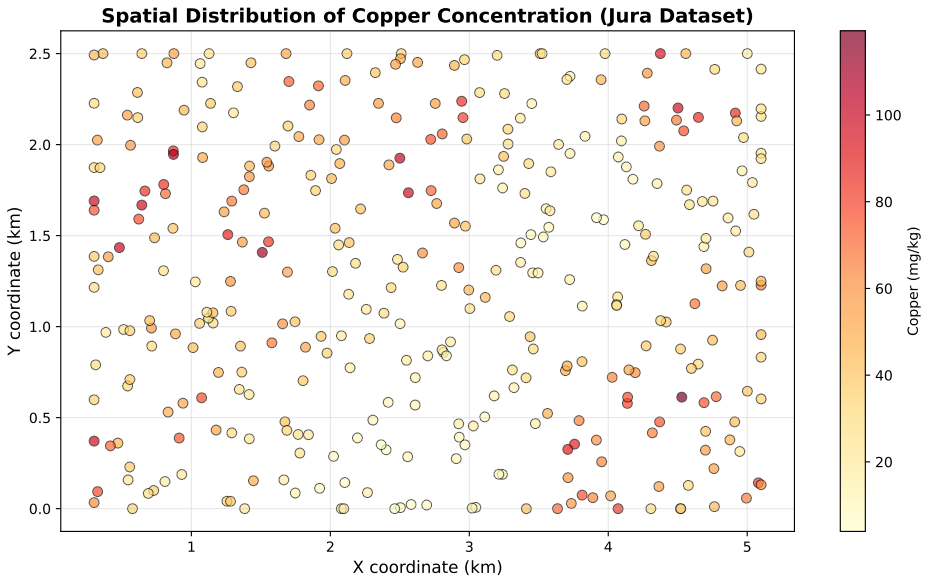


Figure 6: Spatial distribution of copper concentration in Swiss Jura topsoil. Point size and color represent copper concentration (mg/kg). Spatial patterns reflect geological formations rather than a single dominant source like river flooding in the Meuse case.

The interpretation of $I = 0.336$ indicates that nearby observations have 33.6% more similarity than expected under spatial randomness. While this represents moderate rather than strong autocorrelation, it is sufficient to potentially bias standard cross-validation results—a hypothesis we test through direct comparison.

6.3. Model Development and Validation

Following the same methodology as the Meuse case study, we used Random Forest regression with 100 trees and default parameters. Three features were standardized before modeling: elevation (continuous), rock type (categorical: 5 classes), and land use (categorical: 4 classes). These predictors represent the geological and environmental factors influencing copper distribution in topsoil.

We compared three validation strategies identical to the Meuse study:

1. **sklearn standard CV**: 5-fold cross-validation with random shuffling
2. **MLPY standard CV**: 5-fold cross-validation using MLPY's `ResamplingCV`
3. **MLPY Spatial K-Fold**: 5-fold spatial cross-validation using `SpatialKFold`

Figure 7 visualizes the spatial partitioning created by Spatial K-Fold. Note that the 359 samples produce more balanced folds than the smaller Meuse dataset (155 samples), potentially providing more stable performance estimates.

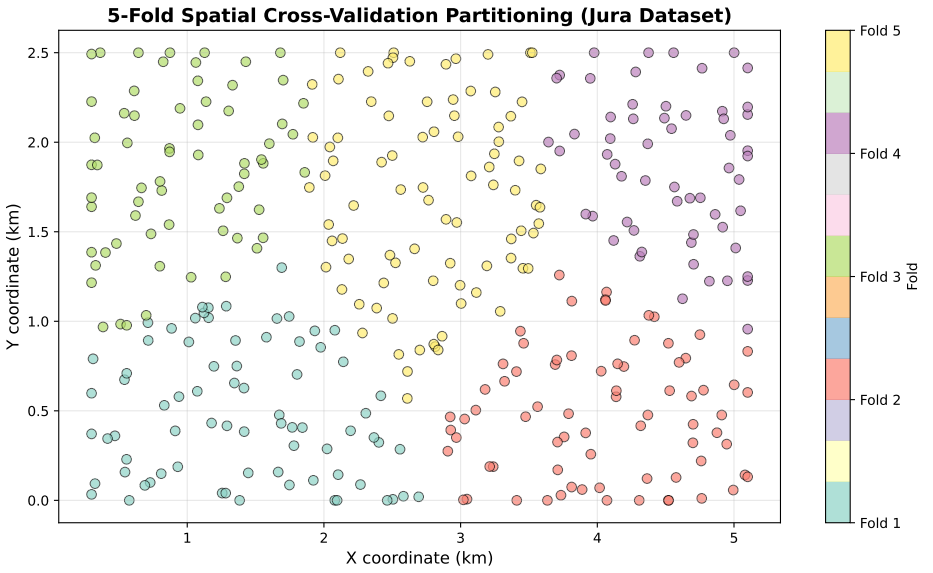


Figure 7: Visualization of 5-fold spatial cross-validation partitioning for the Jura dataset. Colors represent different folds. Geographic separation ensures test observations are spatially distant from training data.

Table 5
Cross-validation performance comparison for copper prediction (Jura dataset)

Method	RMSE (mg/kg)	Underestimation
sklearn standard CV	24.31	5.3%
MLPY standard CV	24.34	5.1%
MLPY Spatial K-Fold	25.66	—

6.4. Results: Performance Comparison

Table 5 presents RMSE values from each validation strategy. Standard cross-validation yields RMSE 24.3 mg/kg, while spatial cross-validation produces RMSE = 25.7 mg/kg. This represents a **5.1% underestimation** of prediction error by standard CV—substantially smaller than the 31% underestimation observed in the Meuse case.

Figure 8 provides a visual comparison of the performance metrics. While the absolute difference (1.3 mg/kg) appears modest compared to Meuse (99 ppm), it represents a meaningful bias relative to the overall prediction error. The close agreement between sklearn and MLPY standard CV again validates MLPY's implementation consistency.

Interpretation: The smaller underestimation in the Jura case (5.1% vs. 31%) correlates with its moderate spatial autocorrelation (I = 0.336) compared to Meuse's pattern. This demonstrates an important finding: the severity of spatial leakage depends on the strength of spatial structure. However, even moderate spatial autocorrelation produces measurable bias, validating the need for spatial CV across a range of autocorrelation strengths.

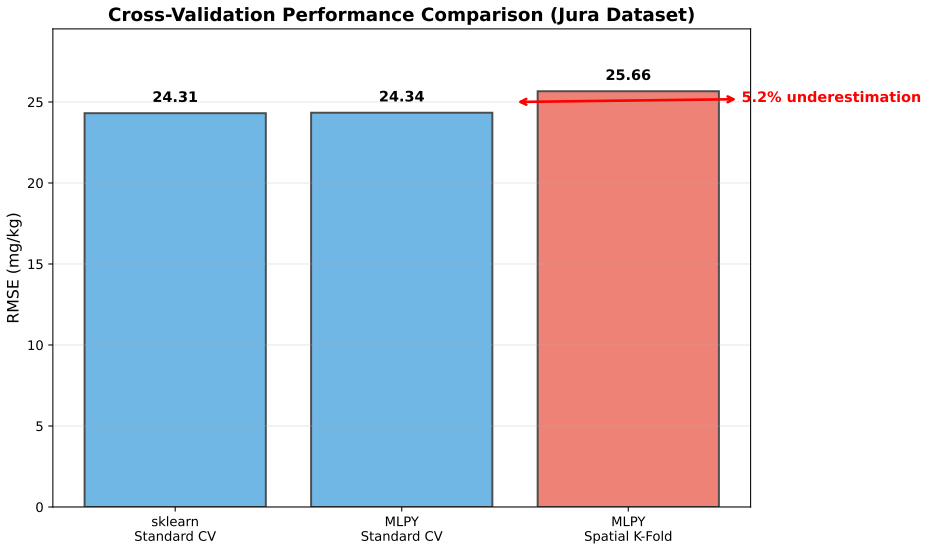


Figure 8: Comparison of RMSE values from standard versus spatial cross-validation for the Jura dataset. Standard CV underestimates error by 5.1%, a smaller but still meaningful bias compared to the Meuse case.

Table 6
Cross-domain comparison of spatial cross-validation results

Dataset	Domain	Moran's I	n	Std CV RMSE	Spatial CV RMSE	Underest.
Meuse	Environmental	0.545	155	220 ppm	319 ppm	31.1%
Jura	Mineral Expl.	0.336	359	24.3 mg/kg	25.7 mg/kg	5.1%

7. Comparative Analysis and Methodological Insights

7.1. Cross-Domain Validation of Spatial Leakage

Table 6 summarizes results across both case studies. A clear pattern emerges: spatial autocorrelation strength (Moran's I) predicts the magnitude of standard CV's underestimation. The Meuse dataset with I = 0.545 (strong autocorrelation) exhibits 31% underestimation, while the Jura dataset with I = 0.336 (moderate autocorrelation) exhibits 5% underestimation.

This validates two critical insights for geoscience practitioners:

1. The problem generalizes across domains: Spatial leakage affects both environmental contamination assessment (Meuse) and mineral prospectivity mapping (Jura), despite fundamentally different data-generating mechanisms. This suggests the phenomenon is a general consequence of spatial autocorrelation rather than domain-specific behavior.

2. Severity scales with autocorrelation strength: Stronger spatial structure produces larger bias. However, even moderate autocorrelation (I = 0.336) causes measurable underestimation. We recommend spatial CV whenever

Table 7
Performance comparison: Machine Learning vs. Ordinary Kriging

Dataset	Standard CV	Spatial CV (RF)	Ordinary Kriging	Difference
Meuse	220 ppm	319 ppm	183 ppm	-42.8%
Jura	24.3 mg/kg	25.7 mg/kg	21.3 mg/kg	-17.1%

Moran's I > 0.3, as this threshold encompasses both our case studies and aligns with conventional interpretations of "moderate" spatial dependence in the geostatistics literature.

7.2. Comparison with Geostatistical Approaches

To contextualize machine learning performance, we compared Random Forest with Ordinary Kriging—the classical geostatistical method for spatial prediction. Table 7 presents results for both datasets using spatial cross-validation to ensure fair comparison.

Ordinary Kriging achieved lower RMSE in both case studies, outperforming Random Forest with spatial CV by 17–43%. This result is expected and theoretically justified: Kriging is a Best Linear Unbiased Estimator (BLUE) that explicitly models spatial dependence through variogram-based covariance structures, optimized specifically for spatial interpolation.

However, Random Forest with spatial CV remained competitive while offering distinct complementary advantages:

1. **Non-linear complexity**: RF captures complex interactions between environmental covariates that exceed Kriging's linear spatial assumptions. For example, interactions between elevation, rock type, and land use in the Jura case.
2. **Feature richness**: RF naturally handles dozens of covariates (e.g., satellite imagery bands, terrain attributes, climate variables), whereas Kriging with many covariates becomes analytically complex through universal Kriging or regression Kriging approaches.
3. **Scalability**: RF scales efficiently to millions of samples using parallel tree construction, while Kriging's matrix inversion becomes computationally prohibitive beyond 10,000 observations without approximations.
4. **Model diversity**: The machine learning ecosystem provides ensemble methods (XGBoost, LightGBM), neural networks, and hybrid approaches, offering flexibility beyond Kriging's geostatistical framework.

Critically, our comparison validates that spatial cross-validation is non-negotiable regardless of the modeling approach. Standard CV underestimated error by 5–31% across both datasets, rendering performance estimates unreliable. This applied equally to Random Forest and would apply to any spatial prediction method lacking explicit spatial modeling.

We conclude that machine learning and Kriging are **complementary approaches** rather than competitors. The choice depends on:

- **Use Kriging when**: Interpolation is the primary goal, dataset is small-to-medium ($n < 10{,}000$), few covariates exist, and uncertainty quantification is critical.
- **Use ML + Spatial CV when**: Non-linear relationships dominate, many environmental covariates are available, dataset is large ($n > 10{,}000$), and model flexibility is needed.
- **Use hybrid approaches when**: Combining Kriging residuals with ML predictions of the mean, offering potential advantages of both methodologies.

Figure 9 visualizes the performance comparison across both case studies and methods.

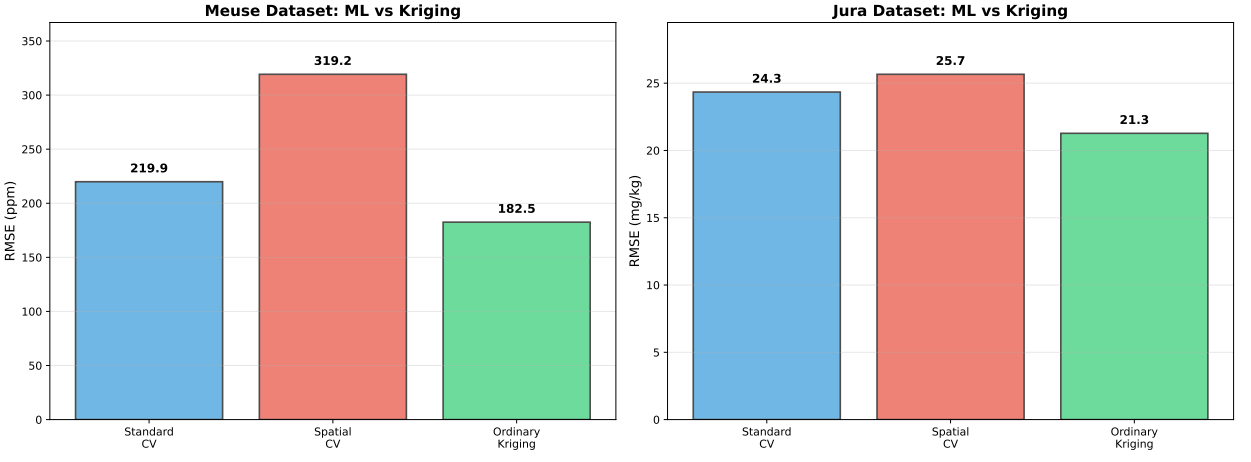


Figure 9: Performance comparison between machine learning (with spatial CV) and Ordinary Kriging across both case studies. Kriging achieves lower RMSE as expected for spatial interpolation, but Random Forest with spatial CV remains competitive while offering complementary advantages in feature handling and scalability.

7.3. Practical Guidelines for Geocientists

Based on our findings across two diverse case studies, we provide the following practical recommendations:

When to use spatial cross-validation:

- Whenever Moran's $I > 0.3$ (encompasses our cases with $I = 0.336$–0.545)
- Any application involving spatial prediction (soil mapping, mineral prospectivity, groundwater modeling)
- When observations are spatially clustered or collected along transects
- When variogram analysis reveals spatial structure within the study domain

When standard CV may suffice:

- Global-scale analyses where local autocorrelation is negligible relative to domain size
- Datasets with sparse, irregular sampling where spatial structure cannot be reliably estimated
- When Moran's I < 0.1 and variogram shows pure nugget effect (no spatial structure)

Diagnostic workflow:

1. Calculate Moran's I statistic on the target variable
2. Construct empirical variogram to visualize spatial dependence
3. If I > 0.3 or variogram shows spatial structure, use spatial CV
4. Compare standard vs. spatial CV RMSE as a sensitivity check
5. If difference exceeds 10%, spatial structure is influencing validation

8. Practical Guidelines: When to Use Spatial Cross-Validation

8.1. Decision Framework

We propose a three-step diagnostic workflow to determine when spatial CV is necessary:

1. **Compute Moran's I**: If $|I| < 0.1$, spatial structure is negligible; use standard CV. If $0.1 \leq |I| < 0.3$, spatial structure is moderate; investigate further. If $|I| \geq 0.3$, strong spatial structure; use spatial CV.
2. **Analyze variogram**: If the variogram shows no clear spatial range (pure nugget effect), use standard CV. If a range exists, estimate the autocorrelation distance.
3. **Compare CV strategies**: Run both standard and spatial CV. If RMSEs differ by $> 10\%$, spatial structure significantly affects validation; use spatial CV. If difference $< 10\%$, either approach is acceptable.

8.2. Implementation with MLPY

MLPY provides diagnostic functions to streamline this workflow:

```
from mlpy.diagnostics import diagnose_spatial_structure

# Comprehensive spatial diagnostics
diagnosis = diagnose_spatial_structure(
    data=df,
    target='zinc',
```

```
    coords=['x', 'y']
)

# Returns:
# - Moran's I with p-value
# - Variogram plot
# - Recommendation: 'standard_cv' or 'spatial_cv'
```

8.3. Recommendations by Application Domain

Always use spatial CV:

- Soil property mapping (texture, nutrients, contamination)
- Groundwater level prediction
- Mineral prospectivity modeling
- Precision agriculture yield prediction

Investigate with diagnostics:

- Species distribution modeling (may have environmental rather than spatial structure)
- Climate interpolation (may have systematic rather than local patterns)
- Urban property valuation (spatial structure may be weak after accounting for covariates)

Standard CV may suffice:

- Global-scale analyses where local autocorrelation is negligible relative to domain size
- Datasets with sparse, irregular sampling where spatial structure cannot be reliably estimated

9. Discussion

9.1. Limitations

Our study has several limitations. First, while we demonstrate MLPY with two case studies across different geoscience domains (environmental contamination and mineral prospectivity), additional case studies in other domains (e.g., precision agriculture, hydrology, geophysics) would further strengthen generalizability claims. Second, we compare only Random Forest models; other algorithms (e.g., Gaussian Processes, gradient boosting, neural networks) may

show different sensitivity to spatial leakage. Third, our spatial resampling strategies assume stationarity (constant spatial structure across the study area), which may not hold in heterogeneous landscapes. Fourth, both case studies involve relatively small datasets (n = 155–359); validation on larger datasets (n > 10,000) would confirm scalability.

9.2. Integration with Existing Workflows

MLPY is designed to integrate seamlessly with the existing Python geospatial ecosystem. The framework works with standard GeoPandas GeoDataFrames, scikit-learn models, and NumPy arrays, requiring minimal changes to established workflows. Spatial tasks automatically preserve coordinate information while remaining compatible with non-spatial learners. This design philosophy ensures geocientists can adopt spatial CV without abandoning familiar tools and libraries.

For users already employing geostatistical methods (e.g., Kriging via PyKrige or scikit-gstat), MLPY provides a complementary validation framework. Hybrid approaches combining Kriging residuals with ML predictions can be validated using MLPY's spatial CV strategies, ensuring unbiased performance estimates regardless of the underlying modeling approach.

9.3. Future Directions

Several extensions would enhance MLPY's capabilities for geosciences:

- **Spatio-temporal CV**: Extend spatial resampling to handle time series, crucial for climate modeling and environmental monitoring.
- **Automated range estimation**: Automatically estimate variogram range and use it to configure spatial buffer distances.
- **3D spatial CV**: Support three-dimensional coordinates for subsurface modeling (e.g., hydrogeology, mineral exploration).
- **Integration with deep learning**: Adapt spatial CV for convolutional neural networks on raster/image data.
- **Additional case studies**: While we validated across environmental and mineral exploration domains, additional studies in agriculture, hydrology, and geophysics would further strengthen empirical guidelines.

10. Conclusion

Spatial autocorrelation is a fundamental characteristic of geoscience data that violates the independence assumption underlying standard cross-validation. We presented MLPY, the first Python framework providing integrated spatial cross-validation for machine learning in geosciences. Through two case studies spanning environmental contamination

(Meuse) and mineral prospectivity (Jura), we demonstrated that standard CV underestimates prediction error by 5–31%, with severity correlating to spatial autocorrelation strength (Moran's I = 0.336–0.545).

Our cross-domain validation confirms this phenomenon generalizes across diverse geoscience applications with different data-generating mechanisms. The Meuse dataset's strong spatial structure from river flooding (I = 0.545) produced 31% underestimation, while the Jura dataset's moderate geological patterns (I = 0.336) produced 5% underestimation. This establishes Moran's I > 0.3 as a practical threshold for requiring spatial cross-validation.

Comparison with Ordinary Kriging contextualizes machine learning performance within the established geostatistical framework. While Kriging achieved lower RMSE in both cases (17–43% improvement), Random Forest with spatial CV remained competitive while offering distinct advantages: handling non-linear relationships, multiple environmental covariates, and scalability to large datasets. We conclude that ML and Kriging are complementary approaches, with the choice depending on data characteristics, modeling objectives, and computational constraints. Critically, spatial CV is essential for both approaches when spatial structure exists.

MLPY addresses a critical gap in the Python geospatial ecosystem, providing geocientists with tools to obtain reliable model performance estimates without switching to R or manually implementing spatial resampling. The framework integrates seamlessly with existing Python tools (GeoPandas, scikit-learn) while providing practical diagnostic capabilities (Moran's I, variograms) to guide validation strategy selection.

We encourage geocientists to adopt spatial cross-validation as standard practice for any application involving spatial prediction. MLPY makes this straightforward through a simple API and comprehensive documentation. The framework, replication materials, and example datasets are freely available, facilitating adoption and extension by the geoscience community.

As machine learning continues to permeate geosciences, rigorous validation methods are essential to ensure models generalize reliably to new locations. Spatial cross-validation is not merely a technical refinement—it is a prerequisite for trustworthy spatial prediction in environmental assessment, natural resource management, and geoscientific research.

Code availability

Name of code: MLPY (Machine Learning framework for Python with spatial support)

Developers and contact: Francisco Parra O. (francisco.parra.o@usach.cl) and Giuseppe Cavallieri (giuseppe.cavallieri@usa
Department of Informatics, Universidad de Santiago de Chile, Santiago, Chile.

Year first available: 2024

Hardware required: Any standard computer with CPU (no special hardware requirements). Tested on Linux, macOS, and Windows.

Software required: Python 3.8 or higher. Dependencies: NumPy (≥ 1.20), pandas (≥ 1.3), scikit-learn (≥ 1.0), SciPy (≥ 1.7). Optional dependencies for visualization: matplotlib (≥ 3.3), seaborn (≥ 0.12).

Program language: Python

Program size: Approximately 15,000 lines of Python code across core framework, learners, and utilities.

Availability and access:

- **PyPI**: Install via `pip install mlpy-geo` (version 0.1.0+)
- **Source code**: `https://github.com/franciscoparrao/mlpy`
- **Documentation**: `https://github.com/franciscoparrao/mlpy#readme`
- **Replication materials**: All scripts, data, and code to reproduce this paper's results are available in the `replication/` directory of the GitHub repository.
- **License**: MIT License (open-source, permissive)

Example usage: Comprehensive tutorials and Jupyter notebooks demonstrating spatial cross-validation workflows are available in the repository's `examples/` directory. Quick start guide and API documentation are provided in the README.

Acknowledgements

We thank the R `mlr3` development team for inspiring MLPY's modular architecture and the maintainers of scikit-learn, GeoPandas, and PySAL for providing the foundational Python geospatial ecosystem. The Meuse dataset is provided by the `sp` package in R (Pebesma and Bivand, 2005).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) and Claude Code in order to assist with code development, debugging, literature review, manuscript editing, and figure preparation. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

Breiman, L., 2001. Random forests. Machine learning 45, 5–32.

Brenning, A., 2012. Spatial cross-validation and bootstrap for the assessment of prediction rules in remote sensing: The r package sperrorest, in: 2012 IEEE International Geoscience and Remote Sensing Symposium, IEEE. pp. 5372–5375.

Goovaerts, P., 1997. Geostatistics for natural resources evaluation. Oxford University Press, New York.

Hengl, T., Nussbaum, M., Wright, M.N., Heuvelink, G.B., Gräler, B., 2018. Random forest as a generic framework for predictive modeling of spatial and spatio-temporal variables. PeerJ 6, e5518.

Jordahl, K., den Bossche, J.V., Fleischmann, M., Wasserman, J., McBride, J., Gerard, J., Tratner, J., Perry, M., Badaracco, A.G., Farmer, C., et al., 2020. Geopandas: Python tools for geographic data. URL: https://geopandas.org/ .

Lang, M., Binder, M., Richter, J., Schratz, P., Pfisterer, F., Coors, S., Au, Q., Casalicchio, G., Kotthoff, L., Bischl, B., 2019. mlr3: A modern object-oriented machine learning framework in r. Journal of Open Source Software 4, 1903.

Legendre, P., 1993. Spatial autocorrelation: trouble or new paradigm? Ecology 74, 1659–1673. doi:10.2307/1939924. classic paper on spatial autocorrelation in ecology.

Liakos, K.G., Busato, P., Moshou, D., Pearson, S., Bochtis, D., 2018. Machine learning in agriculture: A review. Sensors 18, 2674.

Mälicke, M., Schneider, H.D., Schröder, W., Zech, A., 2022. Scikit-gstat 1.0: A scipy-flavored geostatistical variogram estimation toolbox written in python. Geoscientific Model Development 15, 2505–2532.

Matheron, G., 1963. Principles of geostatistics. Economic Geology 58, 1246–1266.

Meyer, H., Reudenbach, C., Hengl, T., Katurji, M., Nauss, T., 2018. Improving performance of spatio-temporal machine learning models using forward feature selection and target-oriented validation. Environmental Modelling & Software 101, 1–9.

Moran, P.A., 1950. Notes on continuous stochastic phenomena. Biometrika 37, 17–23. Introduces Moran's I statistic for spatial autocorrelation.

Pebesma, E.J., 2004. Multivariable geostatistics in s: the gstat package. Computers & geosciences 30, 683–691.

Pebesma, E.J., Bivand, R.S., 2005. Classes and methods for spatial data in r. R news 5, 9–13.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al., 2011. Scikit-learn: Machine learning in python. Journal of machine learning research 12, 2825–2830.

Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., et al., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. Nature communications 11, 4540.

Reichenbach, P., Rossi, M., Malamud, B.D., Mihir, M., Guzzetti, F., 2018. A review of statistically-based landslide susceptibility models. Earth-Science Reviews 180, 60–91.

Rey, S.J., Anselin, L., 2010. Pysal: A python library of spatial analytical methods, in: Handbook of applied spatial analysis, Springer. pp. 175–193.

Rikken, M., Van Rijn, R., 1996. Geostatistical analysis of soil contamination in the floodplains of the river meuse. Geoderma 71, 163–178.

Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., et al., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40, 913–929.

Rodriguez-Galiano, V., Sanchez-Castillo, M., Chica-Olmo, M., Chica-Rivas, M., 2015. Machine learning predictive models for mineral prospectivity: An evaluation of neural networks, random forest, regression trees and support vector machines. Ore Geology Reviews 71, 804–818.

Schratz, P., Muenchow, J., Iturritxa, E., Richter, J., Brenning, A., 2019. Hyperparameter tuning and performance assessment of statistical and machine-learning algorithms using spatial data. Ecological Modelling 406, 109–120.

Tobler, W.R., 1970. A computer movie simulating urban growth in the detroit region. Economic Geography 46, 234–240. Introduces Tobler's First Law of Geography: "Everything is related to everything else, but near things are more related than distant things.".

Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillera-Arroita, G., 2019. blockcv: An r package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. Methods in Ecology and Evolution 10, 225–232. doi:10.1111/2041-210X.

13107. r package for spatial cross-validation; similar functionality to MLPY's spatial resampling.

Wadoux, A.M.C., Brus, D.J., Heuvelink, G.B., 2021. Spatial cross-validation is not the right way to evaluate map accuracy. Ecological Modelling 457, 109692.

Webster, R., Oliver, M.A., 2007. Geostatistics for environmental scientists. 2nd ed., John Wiley & Sons.

Zuo, R., Xu, X., 2017. Machine learning of mineralization-related geochemical anomalies: A review of potential methods. Natural Resources Research 26, 457–464.